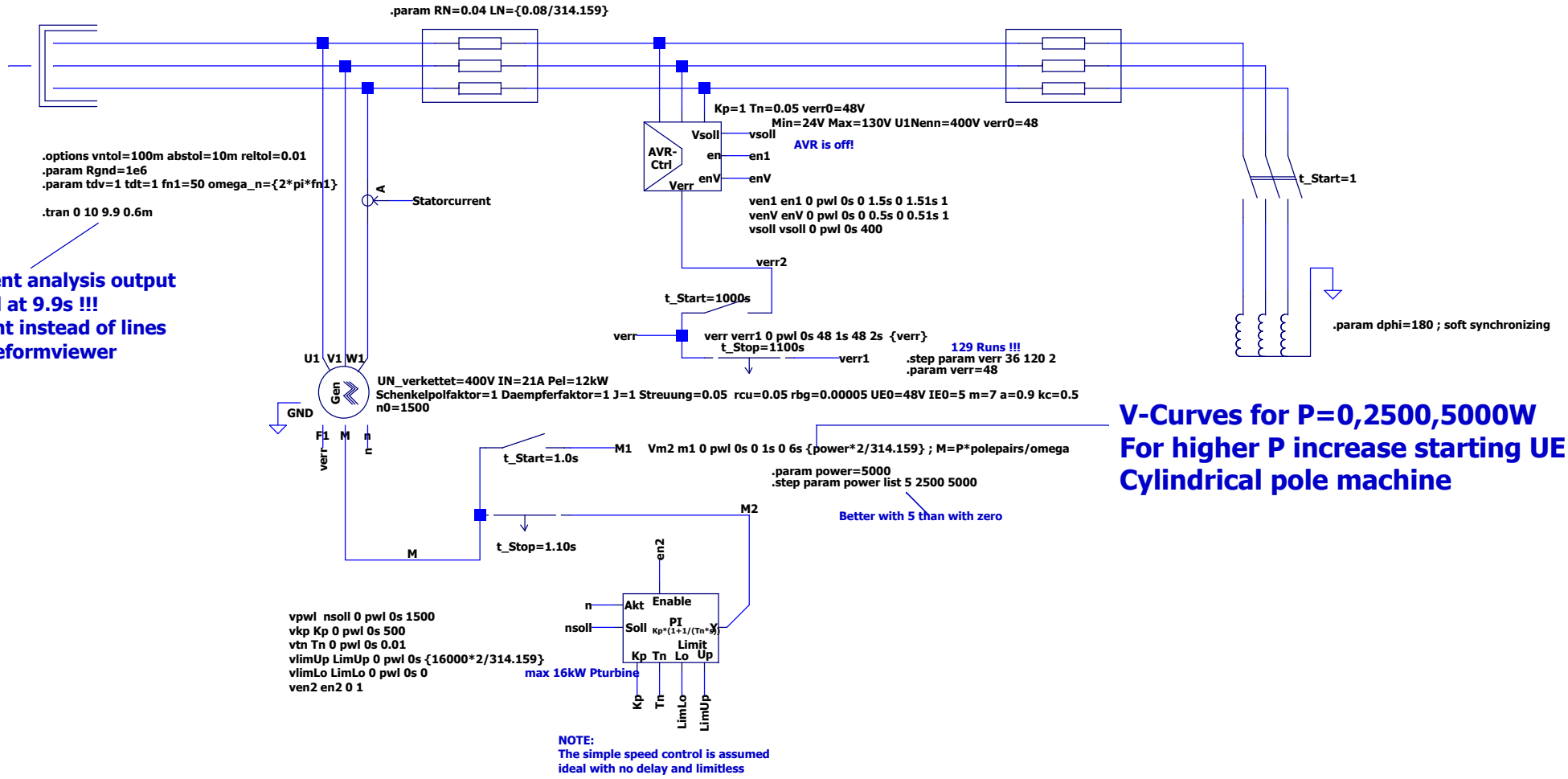
.param RN=0.04 LN={0.08/314.159}
.options vntol=100m abstol=10m reltol=0.01
.param Rgnd=1e6
.param tdv=1 tdt=1 fn1=50 omega_n={2*pi*fn1}
.tran 0 10 9.9 0.6m
Transient analysis output
Started at 9.9s !!!
for point instead of lines
in waveformviewer
A
Statorcurrent
Kp=1 Tn=0.05 verr0=48V
Min=24V Max=130V U1Nenn=400V verr0=48
AVR-Ctrl
Vsoll
en
Verr
enV
vsoll
en1
enV
AVR is off!
ven1 en1 0 pwl 0s 0 1.5s 0 1.51s 1
venV enV 0 pwl 0s 0 0.5s 0 0.51s 1
vsoll vsoll 0 pwl 0s 400
verr2
t_Start=1000s
verr
verr verr1 0 pwl 0s 48 1s 48 2s {verr}
t_Stop=1100s
verr1
129 Runs !!!
.step param verr 36 120 2
.param verr=48
t_Start=1
.param dphi=180 ; soft synchronizing
U1 V1 W1
Gen
UN_verkettet=400V IN=21A Pel=12kW
Schenkelpolfaktor=1 Daempferfaktor=1 J=1 Streuung=0.05 rcu=0.05 rbg=0.00005 UE0=48V IE0=5 m=7 a=0.9 kc=0.5
n0=1500
GND
F1 M n
verr
V-Curves for P=0,2500,5000W
For higher P increase starting UE (>48V)
Cylindrical pole machine
t_Start=1.0s
M1
Vm2 m1 0 pwl 0s 0 1s 0 6s {power*2/314.159} ; M=P*polepairs/omega
.param power=5000
.step param power list 5 2500 5000
Better with 5 than with zero
M2
t_Stop=1.10s
M
en2
n
Akt
Enable
nsoll
Soll
PI
Kp*(1+1/(Tn*s))
Y
Limit
Kp Tn Lo Up
vpwl nsoll 0 pwl 0s 1500
vkp Kp 0 pwl 0s 500
vtn Tn 0 pwl 0s 0.01
vlimUp LimUp 0 pwl 0s {16000*2/314.159}
vlimLo LimLo 0 pwl 0s 0
ven2 en2 0 1
max 16kW Pturbine
Kp
Tn
LimLo
LimUp
NOTE:
The simple speed control is assumed
ideal with no delay and limitless